

REVIEW

Conditional Probability: $P(A | B) = \frac{P(A \cap B)}{P(B)}$

Multiplication Rule: $P(A \cap B) = P(A)P(B | A) = P(B)P(A | B)$

Law of Total Probability: $P(A) = \sum_{\forall i} P(B_i)P(A | B_i)$

Bayes' Theorem: $P(B_i | A) = \frac{P(B_i)P(A | B_i)}{\sum_{j=1}^k P(B_j)P(A | B_j)}$

Note: Conditional probabilities are probabilities.

$$P(A \cap B | C) = P(A | C)P(B | A \cap C)$$

INDEPENDENCE

Learning that B has occurred doesn't change probability of A .

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Ex: Rolling two fair dice.

A : Red die is a 1, B : Blue die is a 2.

$$P(A \cap B) = \frac{1}{36} \text{ and } P(A)P(B) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$



INDEPENDENCE

Ex: Flipping two coins. First coin c_1 flipped, then coin c_2 .

c_1 : Fair coin c_2 : If c_1 is H , flip a fair coin

o.w. flip unfair coin with $P(H) = 0.2$

A : Coin c_1 is H , B : Coin c_2 is H . Are A, B independent?

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$\Rightarrow P(B | A) \neq P(B) \Rightarrow$ not independent.

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c_1 : coin with $P(H) = r$ c_2 : If c_1 is H , flip a coin with $P(H) = p$
o.w. flip a coin with $P(H) = q$

A : Coin c_1 is H , B : Coin c_2 is H .

For $r \neq 1$, what q (in terms of p, r) makes A, B independent?

$q = p$ makes them independent.

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$\Rightarrow P(B | A) \neq P(B) \Rightarrow$ not independent. What is $P(A \cap B)$?

INDEPENDENCE OF SEVERAL EVENTS

$A_1, A_2, \dots, A_n$ are independent if $\forall$ subsets $A_{i_1}, A_{i_2}, \dots, A_{i_j}$,

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_j}) = \prod_{1 \leq k \leq j} P(A_{i_k})$$

INDEPENDENCE OF SEVERAL EVENTS

Ex: The house or the player?

- 4 dice are rolled:
House wins if at least one die is a 6,
otherwise player wins.
- What is the probability that the house wins?



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- Multiple ways:



Events on the dice: A : Die-1 is a 6, B : Die-2 is a 6, ...

$$P(A \cup B \cup C \cup D) = P(A) + P(B) + \dots - P(A \cap B) - P(A \cap C) \dots + P(A \cap B \cap C) + \dots - P(A \cap B \cap C \cap D)$$

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Events on # of 6s: *A*: exactly one die 6, *B*: exactly two dice 6s, ...

$$\binom{4}{1} 5^3/6^4 + \binom{4}{2} 5^2/6^4 + \binom{4}{3} 5^1/6^4 + \binom{4}{4} 5^0/6^4$$

INDEPENDENCE OF SEVERAL EVENTS

Ex: The house or the player?

- 4 dice are rolled:
House wins if at least one die is a 6,
otherwise player wins.
- What is the probability that the house wins?
- Much easier way: Think of the complement

H : At least one die is a 6 $\Rightarrow H^c$: No die is a 6.

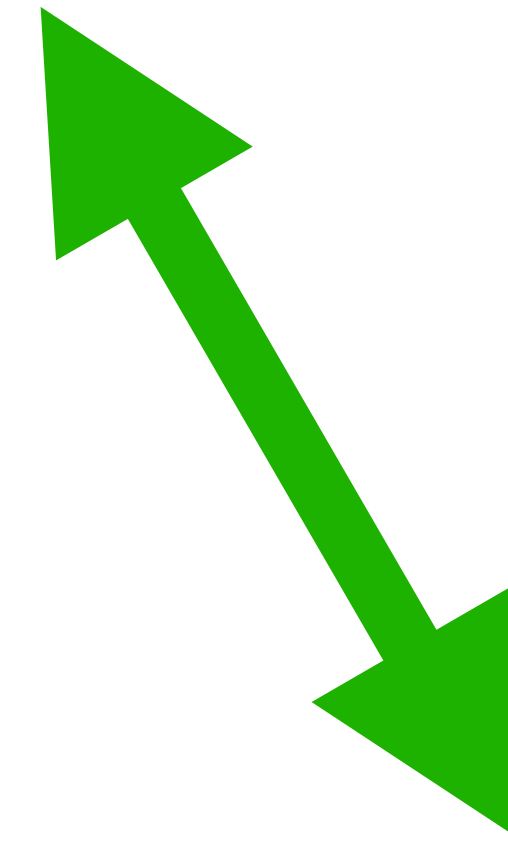
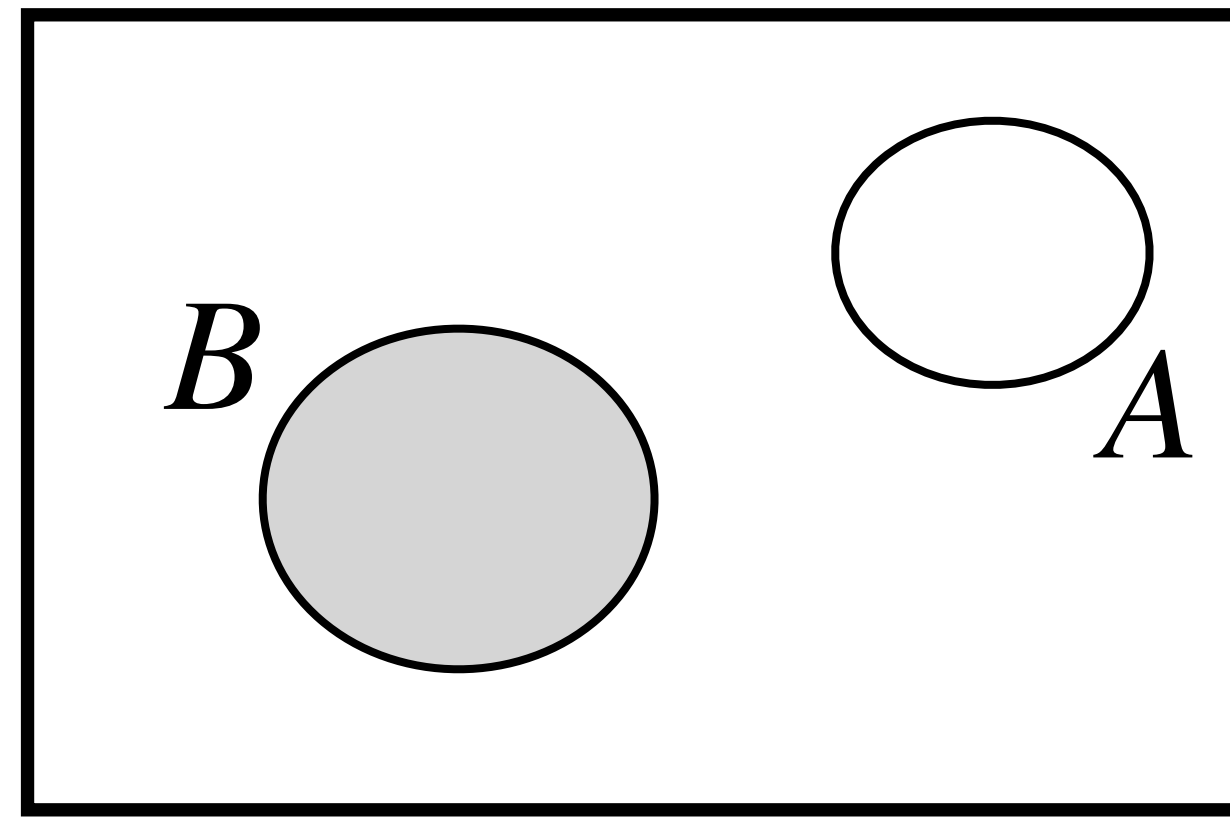
$$P(H) = 1 - P(H^c) = 1 - (5/6)^4 = 0.518$$

Take-home message: The house always wins!



DISJOINT EVENTS VS INDEPENDENT EVENTS

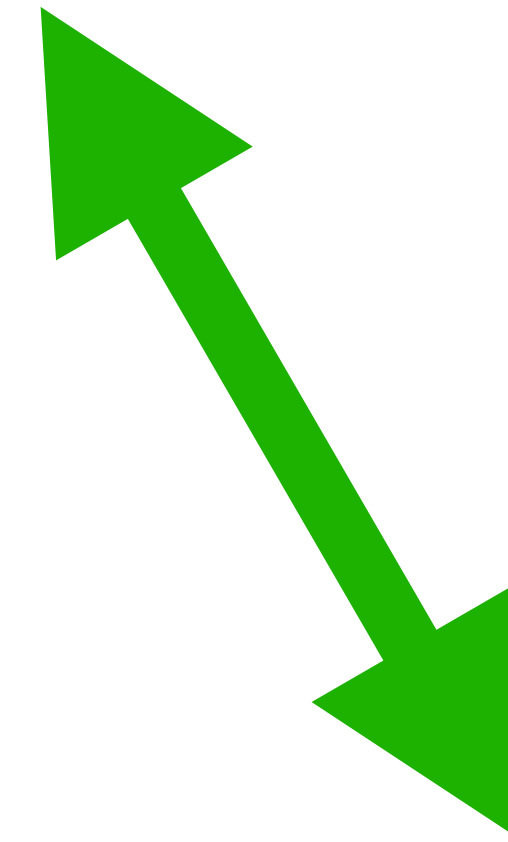
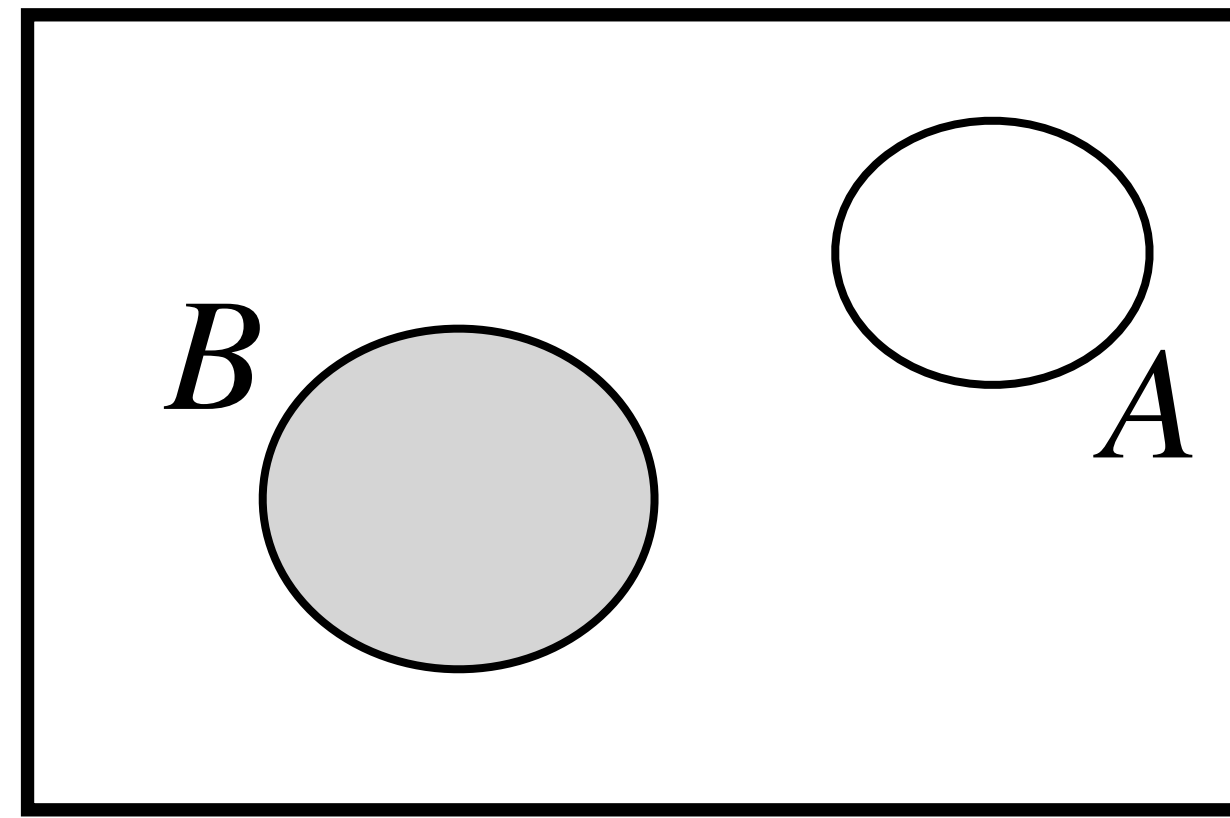
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How about the other direction?

Does $A \cap B \neq \emptyset$ imply A, B independent? **NO!**

RANDOM VARIABLES AND DISTRIBUTIONS

- Readings for the next few lectures:

Ch 7 (Watkins)

Ch 2 (Wasserman, for more formal treatment)

RANDOM VARIABLES

- A real-valued function defined on Ω is a random variable.

Examples:

X : # of heads when a coin is tossed 10 times
gives an integer in range $[0,10]$.

$Y = 10 - X$ also a random variable (# of tails)

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X : the sum of the values of two fair dice
gives an integer in range $[2,12]$.

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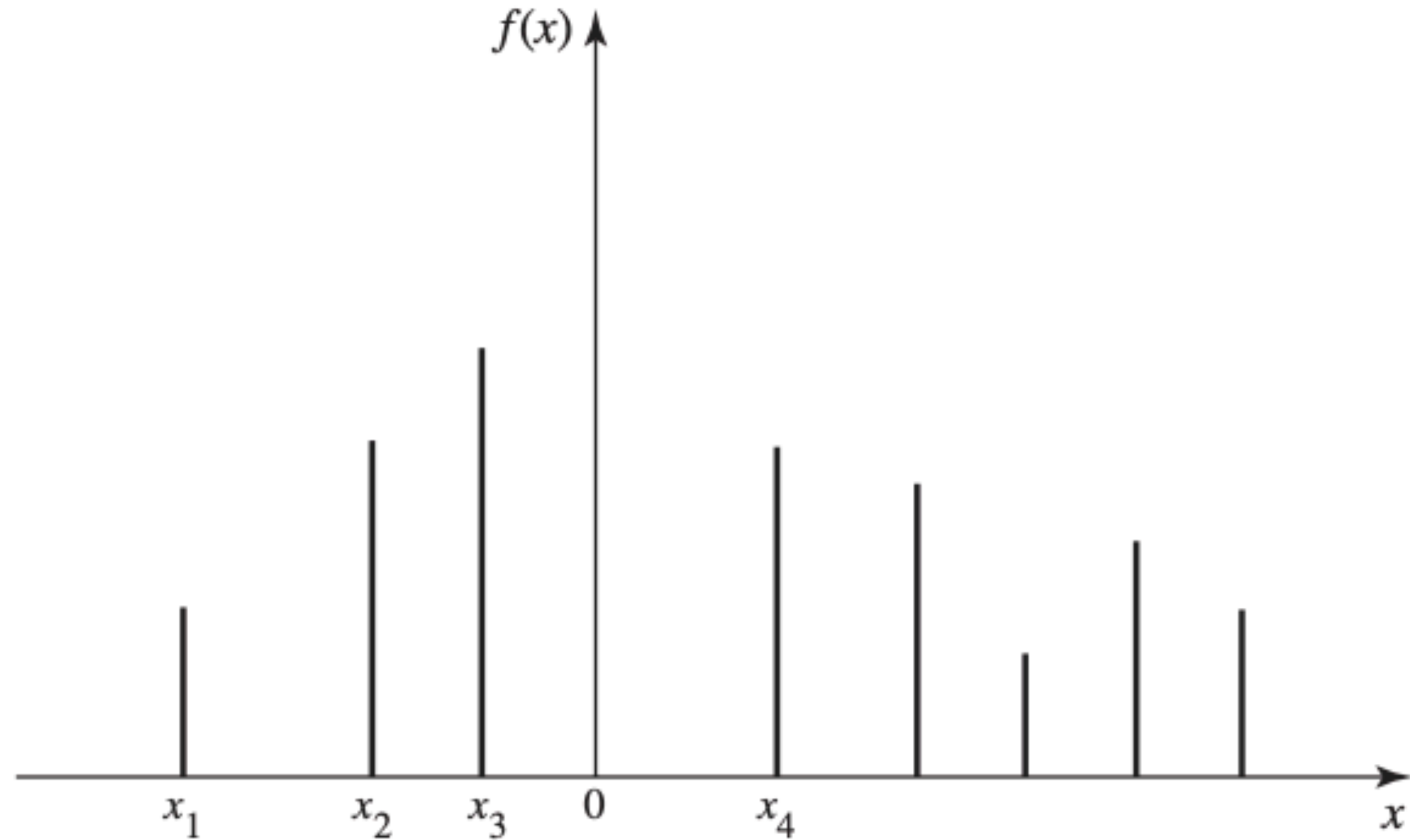
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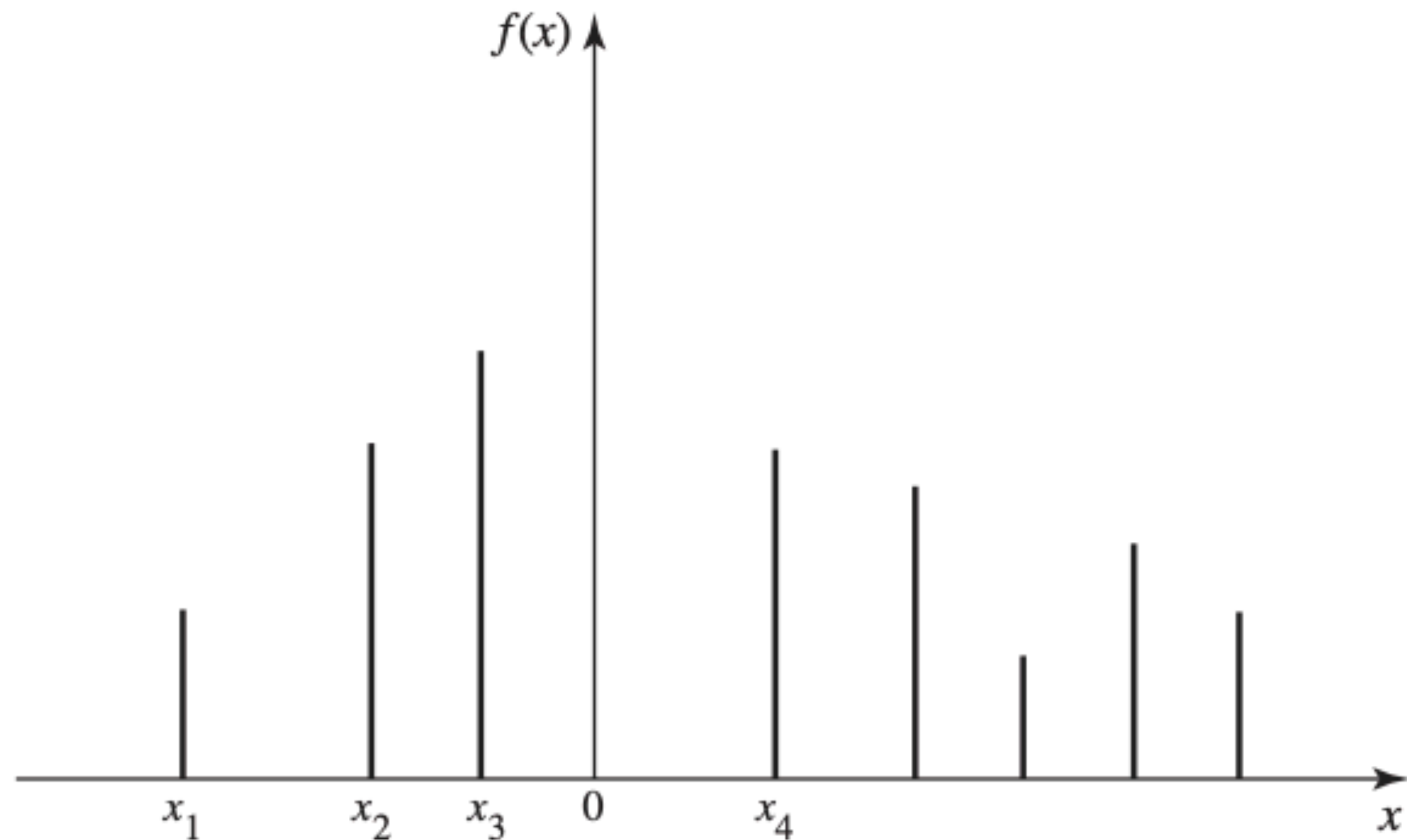
X : height of a person randomly selected from a
population

DISCRETE DISTRIBUTIONS AND RANDOM VARIABLES



- If $\exists$ finite (or countably infinite) values X can take on,
 $\Rightarrow X$ discrete random variable.

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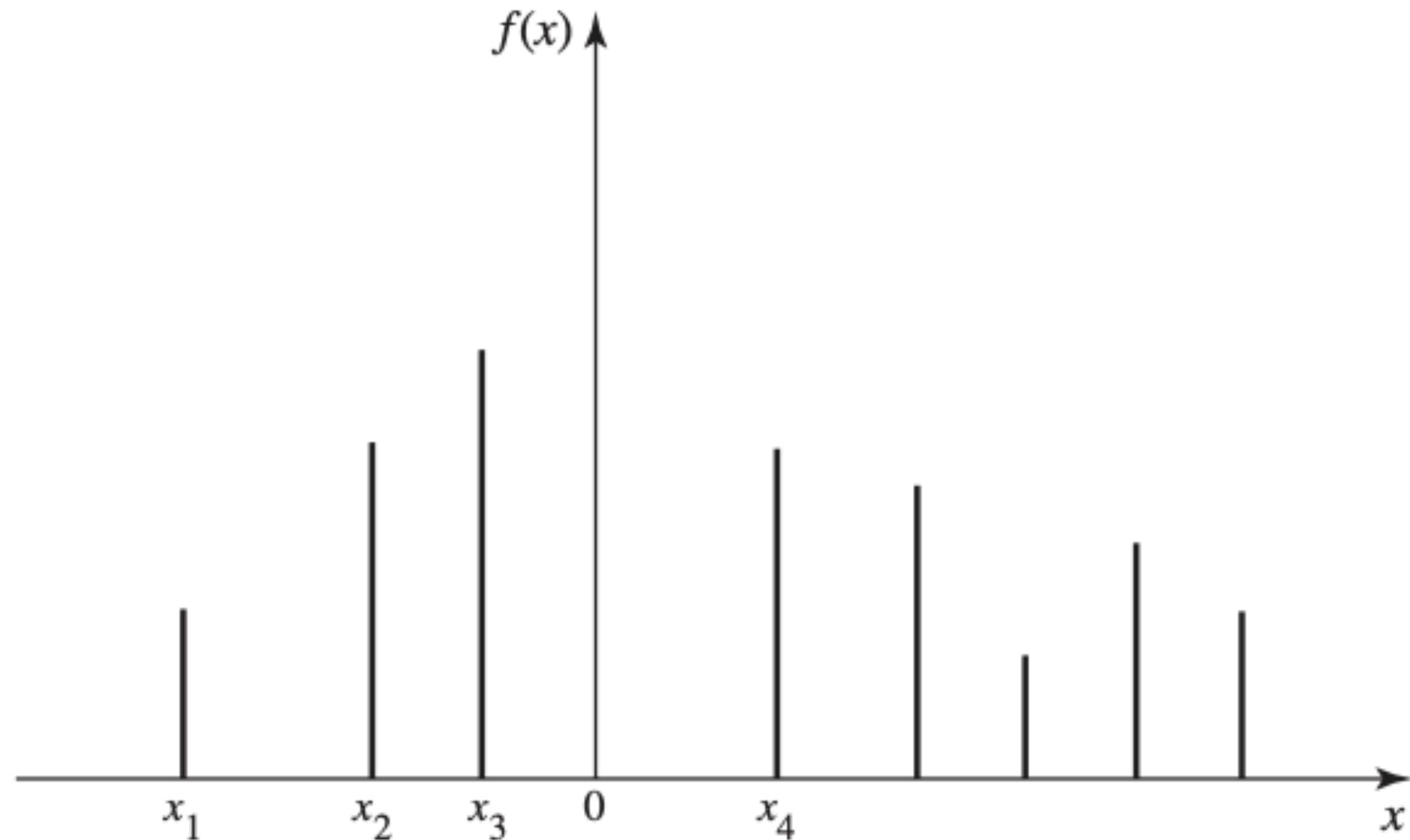


- If $\exists$ finite (or countably infinite) values X can take on,
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$f(x) = P(X = x) \quad \forall x \in \mathbb{R}$ is the probability mass function of X

$\{x : f(x) > 0\}$ is the support of X .

DISCRETE DISTRIBUTIONS AND RANDOM VARIABLES



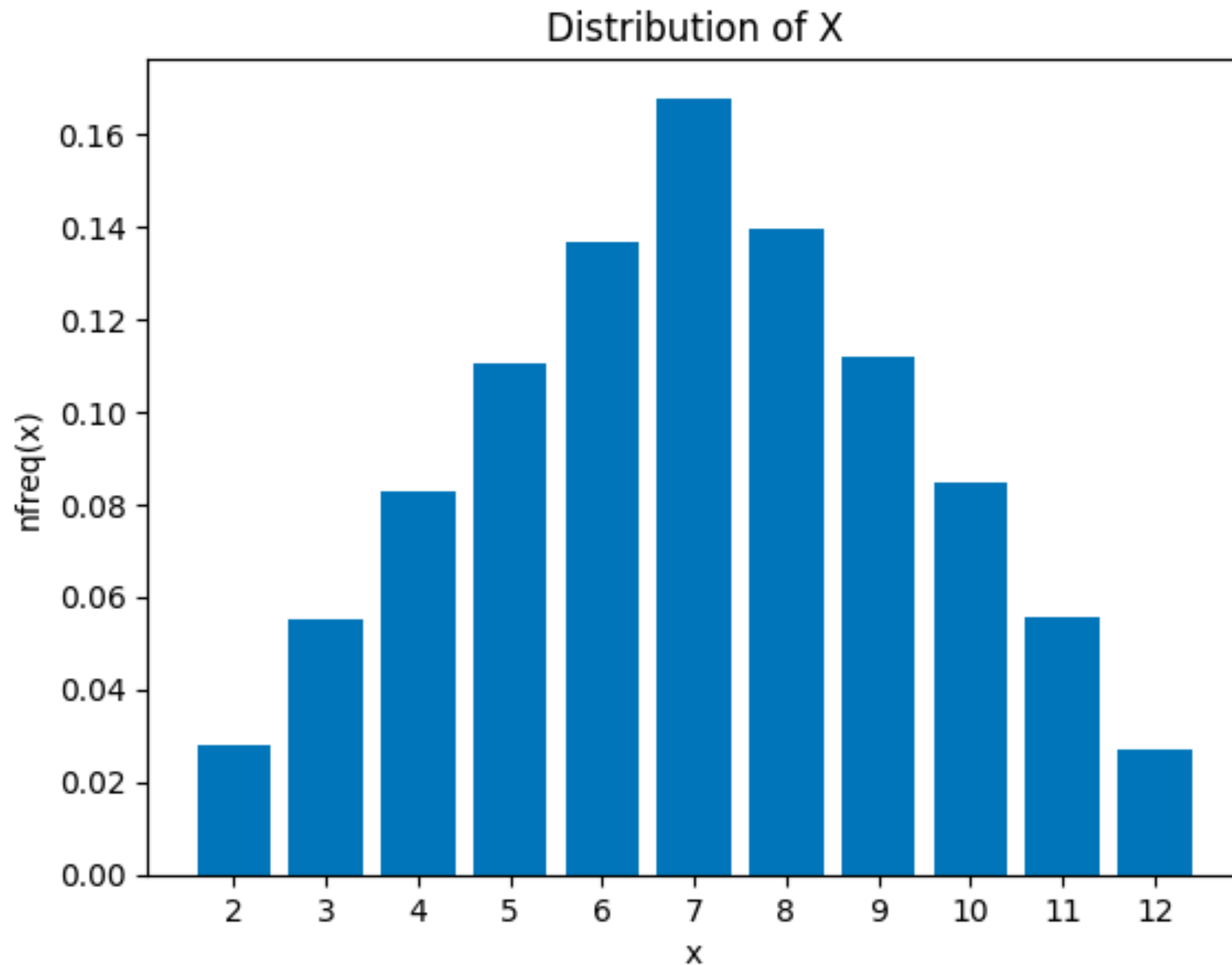
- For any pmf:

- $f(x) \geq 0$

- $\sum_x f(x) = 1$

Note that for an interval: $P(x_2 \leq X \leq x_4) = \sum_{x \in \{x_2, x_3, x_4\}} f(x)$

PMF VS FREQUENCIES VIA SIMULATIONS



X : the sum of two fair dice

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
np.random.seed(5)
N=100_000
dice1 = np.random.randint(1,7,size=N)
dice2 = np.random.randint(1,7,size=N)
dicepairs= [(dice1[i], dice2[i]) for i in range(N)]
```

Find # of dice pairs with sum $X = x$.

Normalize & plot.

COMPLETE NOTEBOOK:

<https://colab.research.google.com/drive/1CJps2wxNUDqswSosaGaPb2Km4URcdqho?usp=sharing>

CODING PROBLEMS

Note: Python code in notebooks. Two alternatives:

- **Colab:** hosted notebooks within Google Drive.
 - No installation
 - <https://colab.research.google.com/>
- **Run notebook on your own machine:**
 - Install Anaconda
 - Use JupyterLab or Jupyter Notebook

Reading: Python for Data Analysis, Ch 4: NumPy Basics

DISCRETE DISTRIBUTIONS AND RANDOM VARIABLES

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Probability mass function of X :

$$f(x) = \begin{cases} \binom{10}{x} \frac{1}{2^{10}} & \text{if } x=0, 1, \dots, 10 \\ 0 & \text{otherwise} \end{cases}$$

and $\{0, 1, \dots, 10\}$ is the support of X . (Binomial distribution)

DISCRETE DISTRIBUTIONS AND RANDOM VARIABLES

Ex: Class with B boys and G girls.

n selected randomly **without replacement**.

X : # of boys in the selection. **Probability mass function of X :**

$$f(x) = \begin{cases} \frac{\binom{B}{x} \binom{G}{n-x}}{\binom{B+G}{n}} & \text{for } \max(0, n-G) \leq x \leq \min(n, B) \\ 0 & \text{otherwise} \end{cases}$$

(Hypergeometric distribution)